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Vision Transformer–Based Denoising with Masked Autoencoders: Preserving Structural Attributes in Seismic Volumes

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Abstract

Seismic denoising is a critical preprocessing step in exploration geophysics, as random noise and acquisition artifacts frequently obscure subsurface reflectors and faults. Traditional denoising methods—such as median filtering, edge-preserving smoothing (EPS), and polynomial smoothing—can suppress noise but often at the expense of blurring geological features. This work introduces a foundation-model approach: we fine-tune a Vision Transformer masked autoencoder (ViT–MAE), pretrained on large seismic datasets (the Seismic Foundation Model, SFM), for three denoising objectives: (1) removal of 15% random noise ("ViT-Random"), (2) emulation of an edge-preserving smoothing filter ("ViT-EPS"), and (3) emulation of a polynomial smoothing filter ("ViT-Poly"). Training and validation are conducted on a real 3-D field volume of size $271 \times 221 \times 876$ (inlines $\times$ crosslines $\times$ time samples), from which 1,084 nonoverlapping 224×224 patches are extracted. Data are split with 80% (≈ 867 patches) for training and 20% (≈ 217 patches) for validation. All inputs are z-score normalized prior to fine-tuning. We compare fine-tuned ViT models against classical filters using peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and mean squared error (MSE). On validation, ViT-Random achieves PSNR = 17.11 dB, SSIM = 0.5584, and MSE = 0.0201 versus median filtering at 13.63 dB / 0.2609 / 0.0444. ViT-EPS yields PSNR = 30.31 dB, SSIM = 0.9661, and MSE = 0.0010 compared to edge-preserving smoothing at 21.15 dB / 0.7856 / 0.0079. ViT-Poly obtains PSNR = 36.56 dB, SSIM = 0.9955, and MSE = 0.0003 versus polynomial smoothing at 22.41 dB / 0.8458 / 0.0065. Loss curves confirm stable convergence within 100 epochs. On an unseen Norway F3 cube ($458 \times 623 \times 1001$), ViT-Random achieves PSNR = 29.42 dB, SSIM = 0.8853, and MSE = 0.0011—despite no additional retraining—demonstrating strong generalization to real field data. Although classical filters yield higher quantitative scores, ViT-Random preserves more structural detail and textural continuity. Qualitative results show that ViT models suppress noise while preserving reflector continuity, thin-bed stratigraphy, and fault sharpness. These findings highlight the value of pretrained seismic models and demonstrate that fine-tuning on synthetic perturbations—with minimal labeled data—can yield structure-aware denoising models that generalize well to real-world seismic volumes.

Introduction

Seismic reflection data acquisition and processing aim to produce accurate subsurface images for reservoir characterization, fault mapping, and stratigraphic interpretation. However, seismic volumes are often contaminated by both random noise and coherent energy (e.g., multiples, ground roll), which degrade signal-to-noise ratio and obscure subtle reflectors. Effective denoising is therefore essential to enhance seismic clarity before migration, inversion, or attribute analysis.

Classical denoising methods—such as median filtering, structure-oriented smoothing (e.g., edge-preserving smoothing, EPS; Luo et al., 2002), and polynomial fitting (Lu & Lu, 2020)—are widely used but often require manual parameter tuning and may oversmooth geological features. While effective in suppressing specific noise types, they struggle to generalize across surveys or adapt to varying geologic contexts.

Recent advances in deep learning, particularly convolutional neural networks (CNNs; Ronneberger et al., 2015) and self-supervised methods like NOISE2VOID (Krull et al., 2019), offer greater flexibility but are limited by local receptive fields and often require task-specific training. In contrast, foundation models—large pretrained networks designed for downstream adaptation—can learn global patterns from unlabeled data. In computer vision, masked autoencoders (MAEs; He et al., 2022) reconstruct masked image patches to learn contextual features; this approach has been adapted to seismic data by Sheng et al. (2025), who introduced the Seismic Foundation Model (SFM), a ViT–MAE pretrained on over two million seismic patches from 192 volumes.

Motivated by the need to develop flexible, structure-preserving denoising models that do not require extensive labels or hand-tuned parameters, we explore three lightweight fine-tuning strategies atop the pretrained SFM backbone. ViT-Random targets synthetic noise suppression, representing a general-purpose approach that does not rely on prior filter design. ViT-EPS and ViT-Poly, by contrast, are trained to emulate widely accepted classical filters—edge-preserving smoothing (Luo et al., 2002) and polynomial smoothing (Lu & Lu, 2020)—thus enabling interpreters to obtain familiar, explainable outputs while benefiting from the adaptability of deep learning. This design allows us to assess how foundation models can bridge data-driven denoising with domain-standard filtering practices.

Building on this pretrained SFM backbone, we fine-tune three lightweight decoder heads for task-specific denoising: ViT-Random (trained on synthetic random noise), ViT-EPS (to replicate edge-preserving smoothing), and ViT-Poly (to mimic polynomial filtering). Performance is evaluated quantitatively using PSNR, SSIM, and MSE on a real 3-D seismic volume ($271 \times 221 \times 876$) and qualitatively assessed through structural preservation. To assess generalization, ViT-Random is further tested on the Norway F3 cube ($458 \times 623 \times 1001$) without retraining. Our findings show that fine-tuned ViTs offer robust, structure-aware denoising across both synthetic and real-world conditions.

Related Work

This section reviews classical and modern denoising techniques, the progression toward structure-preserving filters, and the recent emergence of foundation models tailored to seismic applications.

Classical Seismic Denoising Filters

Traditional denoising techniques in seismic processing offer varied trade-offs. Median filters, commonly using 3×3 or 5×5 windows, are effective at removing impulse noise but blur reflector edges and degrade spatial continuity in coherent noise scenarios. Structure-oriented filters address this by estimating local dips and smoothing along structural features. For instance, Weber and Hale (1988) introduced early dip-based filters, which were refined by Fehmers and Höcker (2003) through anisotropic diffusion methods. A prominent variant is Luo et al.'s edge-preserving smoothing (EPS; 2002), which leverages plane-wave destruction or semblance analysis to maintain continuity and faults during filtering. Polynomial smoothing

—originating from [Savitzky and Golay \(1964\)](#) and adapted to seismic data by [Lu and Lu \(2020\)](#)—fits low-order polynomials to seismic trace windows. While effective at smoothing high-frequency noise, this method can oversmooth near faults if window sizes or polynomial orders are suboptimal. All classical methods typically require manual parameter tuning and may not generalize well across varying noise types or geological settings.

Deep Learning in Seismic Denoising

Deep learning models have significantly advanced seismic denoising. CNNs—especially U-Net architectures ([Ronneberger et al., 2015](#))—learn mappings from noisy to clean patches when sufficient labeled data are available ([Qiu et al., 2018](#); [Zhang and Ma, 2019](#)). However, their inherently local receptive fields often limit their ability to preserve long-range structural continuity. Self-supervised approaches like NOISE2VOID ([Krull et al., 2019](#)) and Noise2Self ([Batson and Royer, 2019](#)), later adapted for seismic data ([Venugopalan et al., 2021](#)), allow denoising without clean labels but still depend on local context and may fail to maintain global structural fidelity. The Seismic Foundation Model (SFM), introduced by [Sheng et al. \(2025\)](#), addresses this by using a ViT–MAE architecture pretrained on over two million unlabeled seismic patches. The SFM demonstrates strong performance in facies classification, fault detection, interpolation, and denoising, although its ability to emulate classical filters had not been systematically evaluated before this study.

Vision Transformers and Masked Autoencoders

Vision Transformers (ViTs), originally introduced by [Vaswani et al. \(2017\)](#), process images by dividing them into non-overlapping patches, encoding them as tokens, and leveraging multi-head self-attention to capture long-range dependencies—bypassing the constraints of convolutional receptive fields. [He et al. \(2022\)](#) proposed masked autoencoders (MAEs), wherein only a fraction of patches are visible to the encoder; the decoder then reconstructs the full image—enabling rich contextual learning. In seismics, SFM adopts this ViT–MAE paradigm: massive pretraining on unlabeled seismic patches facilitates representation learning of geological structures and noise-invariant patterns, yielding strong generalization in fine-tuned tasks ([Sheng et al., 2025](#)).

Emerging Foundation Models in Geophysics

Beyond SFM, other foundation models are emerging for geophysical applications. Altay [Sansal et al. \(2025\)](#) introduced a scalable 3D seismic ViT–MAE approach trained on 63 global seismic volumes, demonstrating zero-shot generalization for salt interpretation. [ThinkOnward \(2025\)](#) developed an open-source geophysical foundation model using ViT–MAE with trace-based masking, broadening accessibility and adoption. In seismology, Xu [Si et al. \(2023\)](#) explored contrastive, multi-modal pretraining of transformer encoders on seismic spectrum and metadata through SeisCLIP, improving event detection and classification tasks. Moreover, Xinghao [Wang et al. \(2025\)](#) proposed SeisMoLLM, applying cross-modal transfer from LLMs to seismic waveform tokenization, addressing tasks like phase picking and magnitude estimation with high accuracy.

Summary of Gaps

Despite substantial progress, two limitations remain prominent. First, structural preservation: many CNN-based and classical methods smooth out faults and subtle reflectors during noise reduction. ViT–MAE frameworks, with long-range attention, promise better maintenance of geological structure. Second, classical filter emulation: while deep learning methods focus on noise suppression, they rarely replicate established filters like EPS or polynomial smoothing. A ViT model with such emulation capability would offer interpreters familiar output, improved automation, and consistency. To address these, our study fine-tunes three task-specific ViT models—starting from the same SFM encoder—for: (a) random-noise

removal ("ViT-Random"), (b) EPS filter emulation ("ViT-EPS"), and (c) polynomial smoothing emulation ("ViT-Poly"). This unified framework leverages pretrained representations and performs structure-aware denoising across multiple paradigms.

Methodology

This section outlines the complete experimental workflow for fine-tuning Vision Transformer (ViT) models on seismic denoising tasks. It details the data preparation process, model architecture, training configuration, and evaluation setup for generalization.

Data Preparation

Experiments are conducted on a proprietary 3D post-stack seismic volume comprising 271 inlines, 221 crosslines, and 876-time samples. Before training, the full volume undergoes task-specific preprocessing tailored to each denoising task.

For the ViT-Random task, synthetic random noise is injected into the original cube to simulate amplitude corruption while preserving geological structure. The noise is scaled to 15% of each patch's original dynamic range. Specifically, for a clean input patch $x_{i,j}$, the noisy version $\widetilde{x}_{i,j}$ is computed as:

$$\widetilde{x}_{i,j} = x_{i,j} + \alpha \cdot r_{i,j},$$

Where $r_{i,j} \sim N(0,1)$ is a random variable and $\alpha = 0.15 \cdot (\max x - \min x)$ ensures the noise amplitude is 15% of the patch dynamic range. The noisy patch $\widetilde{x}$ serves as the model input, and the corresponding clean patch x is used as the labeled target.

For the ViT-EPS and ViT-Poly tasks, industry standard classical filters are applied to the full cube before slicing. We apply the edge-preserving smoothing method of [Luo et al. \(2002\)](#), configured with an 11-sample window, diffusion factor of 0.8, and coherence threshold of 0.5. This filter enhances reflector continuity while avoiding blurring across structural boundaries. For polynomial smoothing, we adopt the method by [Lu and Lu \(2020\)](#), which applies a third-order polynomial fit over a 21-sample vertical window. This technique suppresses random noise while retaining subtle stratigraphic trends and reflector curvature. For both ViT-EPS and ViT-Poly, the original seismic volume is used as the model input, and the filtered volume (EPS or polynomial) is used as the labeled target. Preprocessing the entire 3D cube prior to slicing preserves the lateral continuity of reflectors across traces, ensuring that each patch retains realistic geological context.

After preprocessing, each version of the cube is sliced along the inline axis to produce 2D crossline–time sections. From each inline, four non-overlapping patches of size 224×224 are extracted, resulting in 1,084 patches. These are split into 867 for training and 217 for validation using 80/20 split, consistently applied across all tasks. Prior to fine-tuning, all patches are z-score normalized using statistics derived from the training set to maintain consistent input scaling.

Model Architecture

All experiments in this study utilize the pretrained Seismic Foundation Model (SFM) developed by Sheng et al. (2023). The SFM is a masked autoencoder based on the ViT-Base architecture, originally trained on a massive corpus of over 2 million seismic patches from 192 surveys spanning diverse geological settings, acquisition parameters, and structural styles. The encoder consists of 12 transformer blocks with 768 hidden dimensions and 12 attention heads, optimized for capturing both local stratigraphy and long-range geologic continuity.

Pretraining follows the masked reconstruction paradigm introduced by [He et al. \(2022\)](#), wherein a subset of input patches is randomly masked and the model learns to reconstruct missing regions based on contextual information. This large-scale self-supervised pretraining equips the encoder with a strong prior over seismic structure, enabling effective adaptation to downstream tasks with minimal task-specific supervision.

Building on this pretrained backbone, we fine-tune the SFM on three supervised denoising tasks—ViT-Random, ViT-EPS, and ViT-Poly—each designed to target a different type of noise or structural refinement. The fine-tuning procedure adapts the pretrained encoder and decoder weights to task-specific objectives while preserving the generalization capacity acquired during pretraining.

Finetuning Strategy

Each ViT variant—ViT-Random, ViT-EPS, and ViT-Poly—is fine-tuned independently using the pretrained Seismic Foundation Model (SFM). Fine-tuning is conducted end-to-end, updating both encoder and decoder parameters to align the model's representations with task-specific targets. Training proceeds for 100 epochs using the AdamW optimizer (Loshchilov and Hutter, 2019), initialized with a learning rate of 1.5×10^{-3} and annealed via a cosine decay schedule. To reduce GPU memory usage and accelerate training, mixed-precision computation is employed. A batch size of 64 is used, distributed across four NVIDIA V100 GPUs. The training objective combines pixel-level accuracy with perceptual structure preservation. Specifically, the total loss is a hybrid of mean squared error (MSE) and multi-scale structural similarity (MS-SSIM; Wang et al., 2004), defined as:

$$\mathcal{L} = (1 - \alpha) \cdot \text{MSE}(x, \hat{x}) + \alpha \cdot \text{MSSSIM}(x, \hat{x}), \quad \alpha = 0.5$$

where x denotes the ground truth patch (clean, EPS-smoothed, or polynomial-smoothed), and $\hat{x}$ is the model prediction. The weighting coefficient $\alpha = 0.5$ was empirically chosen to balance numerical accuracy with structural preservation, reflecting the dual goals of amplitude restoration and reflector continuity.

Inference Setup for Generalization

To assess generalization, we evaluate the ViT-Random model on the publicly available Netherlands Offshore F3 dataset (dGB Earth Sciences, 2020), which was excluded from all training phases. The F3 volume comprises 458 inlines, 623 crosslines, and 1001 time samples, and represents a geologically distinct North Sea environment compared to the training data.

The volume is sliced along the inline direction to produce 224×224 patches, consistent with the training setup. For qualitative comparison, model predictions are benchmarked against outputs from industry-standard filters—edge-preserving smoothing and polynomial smoothing applied directly to the F3 volume. These filters provide consistent reference baselines for evaluating the model's ability to recover clean seismic structure under realistic conditions.

Results

Random-Noise Removal (ViT-Random)

Validation Results. The ViT-Random model was fine-tuned to suppress synthetic random noise and evaluated on 217 validation patches. As shown in Table 1, it outperforms the classical 3×3 median filter across all metrics, yielding higher structural similarity and lower reconstruction error. It reduces mean squared error by over 50% and more than doubles SSIM compared to the baseline, indicating improved reflector continuity and noise suppression without blurring critical features.

Table 1—Quantitative performance of ViT-Random versus median filtering. Metrics are computed against clean reference patches.

Method	PSNR (dB)	SSIM	MSE
Median filter	13.63	0.2609	0.0444
ViT-Random	17.11	0.5584	0.0201

Fig. 1 provides visual validation of these results. Each row shows a denoising example, including (from left to right) the noisy input, clean label, denoised output, and residual map. The top row illustrates the output from ViT-Random, which effectively restores reflector continuity, enhances subtle geologic features, and preserves structural edges with minimal residuals. In contrast, the bottom row shows the result of median filtering, which reduces high-frequency noise but blurs reflector boundaries and suppresses fine structural detail.

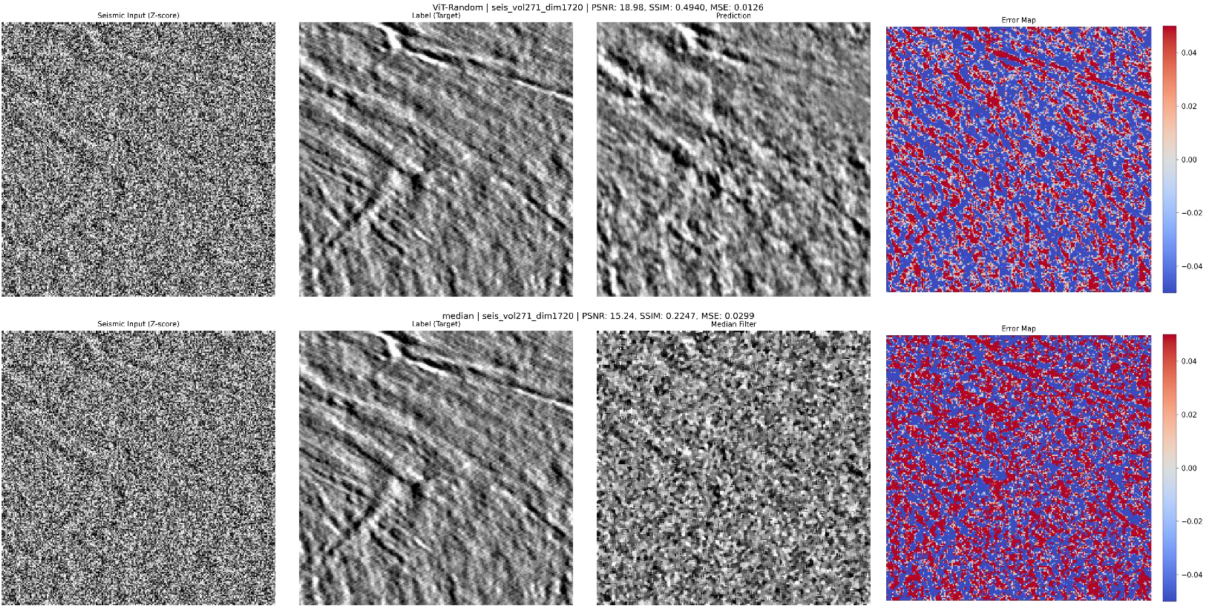


Figure 1—Validation results for random-noise removal. Top row: ViT-Random prediction; bottom row: median filter. Each row shows (left to right) the noisy input, clean label, denoised output, and residual map. ViT-Random recovers reflector continuity and small-scale geologic features with fewer artifacts. Median filtering removes noise but smooths reflectors and introduces broader residuals.

These visual patterns reinforce the quantitative metrics: while both methods reduce noise, the ViT-Random model better distinguishes coherent events from random artifacts. Residual maps confirm that ViT-Random leaves fewer and more spatially localized errors, indicating higher structural fidelity.

Generalization to Norway F3 Dataset. To evaluate generalization, the ViT-Random model was directly applied to the unseen Norway F3 dataset ($458 \times 623 \times 1001$) without any fine-tuning. Its denoising performance was compared against two classical industry-standard filters: the edge-preserving smoothing (EPS) filter and the polynomial smoothing method.

As summarized in Table 2, ViT-Random achieved PSNR = 29.42 dB, SSIM = 0.8853, and MSE = 0.0011. While the EPS and polynomial filters yielded higher quantitative scores, ViT-Random preserved more of the finer textural elements and thin-bed reflectors. The polynomial filter achieved a near-zero MSE but visually exhibited over-smoothing of subtle structures. The EPS filter retained slightly more structural definition than polynomial smoothing but was less effective in preserving local texture compared to ViT-Random.

Table 2—Quantitative performance of ViT-Random against industry-standard filters (EPS and polynomial smoothing) on the unseen Norway F3 dataset.

Method	PSNR (dB)	SSIM	MSE
Vit-Random	29.42	0.8853	0.0011
EPS Filter	33.26	0.9803	0.0005
Polynomial Smoothing	48.81	0.9990	0.0001

Fig. 2 provides side-by-side comparisons of the raw input, ViT-Random output, and the outputs from EPS and polynomial smoothing. The ViT-Random output recovers reflector boundaries and directional continuity while maintaining high-frequency detail. Compared to EPS and polynomial filters, which suppress local variations, the ViT-Random prediction offers a visually balanced compromise between smoothing and structure retention.

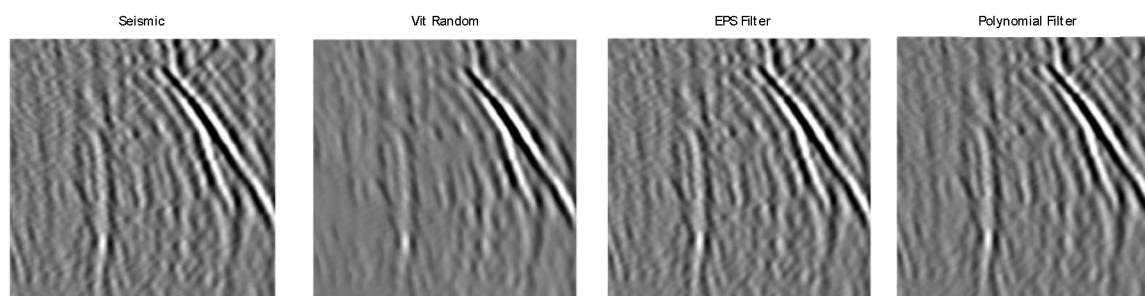


Figure 2—Inference on Norway F3: From left to right — raw input, ViT-Random output, EPS filter output, and polynomial smoothing output. ViT-Random retains reflective continuity and subtle layering, whereas classical filters produce smoother but less textured outputs.

EPS Filter Model (ViT-EPS)

For the EPS task, the model was trained to emulate an edge-preserving smoothing filter using clean seismic data as input and EPS-filtered outputs as labels. Fig. 3 shows three representative validation examples, each presenting the input, EPS-smoothed label, ViT-EPS output, and the residual map. Visually, the predictions closely match the smoothed labels, preserving major reflectors while enhancing subtle features such as slope breaks and minor lobes. The residuals are spatially sparse and low in magnitude, confirming the model's fidelity to the EPS transformation.

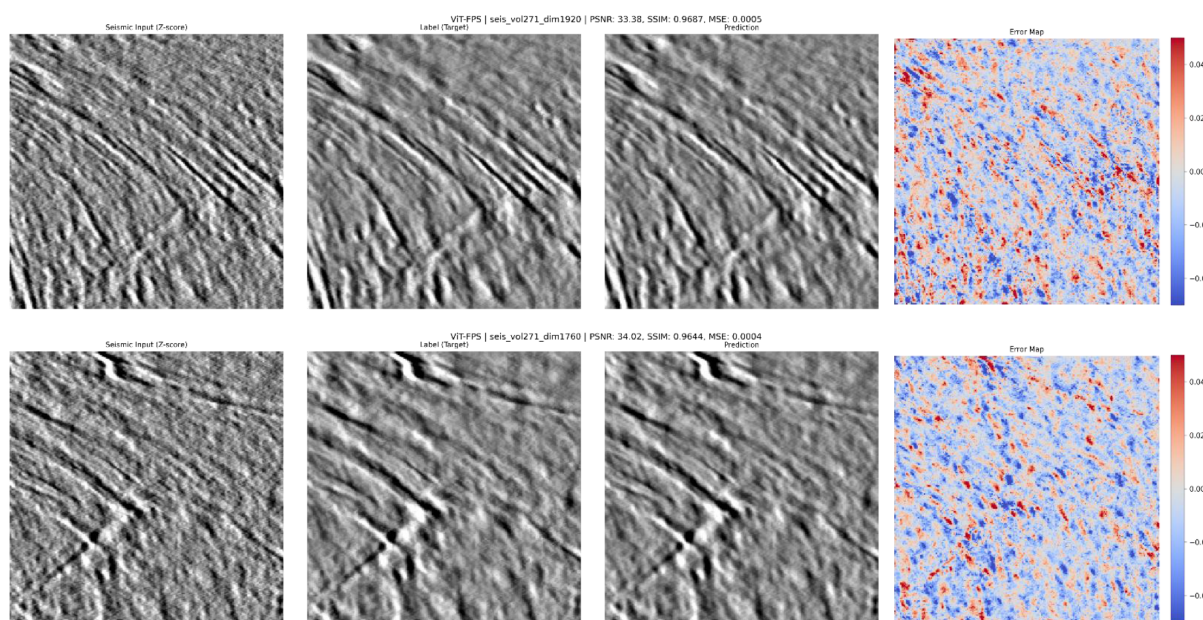


Figure 3—Visual comparison of EPS smoothing. Each row shows a validation patch with (left to right): raw input, EPS label, ViT-EPS output, and residual (ViT-EPS - EPS label). The ViT model captures the smoothing behavior while preserving fine geologic details.

Quantitative results are summarized in Table 3. Compared with the EPS-filtered reference, the ViT-EPS output achieves high structural similarity and low reconstruction error, demonstrating the model's effectiveness in emulating industry-standard smoothing with greater consistency and detail retention.

Table 3—Quantitative performance of ViT-EPS against industry-standard filters EPS on validation data.

Method	PSNR (dB)	SSIM	MSE
EPS filter	21.15	0.7856	0.0079
Vit-EPS	30.31	0.9661	0.0010

Polynomial Smoothing Model (ViT-Poly)

ViT-Poly targets the output of the edge-preserving polynomial smoothing filter introduced by Lu and Lu (2020). Fig. 4 presents qualitative results from three representative validation patches, showing the input, polynomial-smoothed label, ViT-Poly output, and the residual maps.

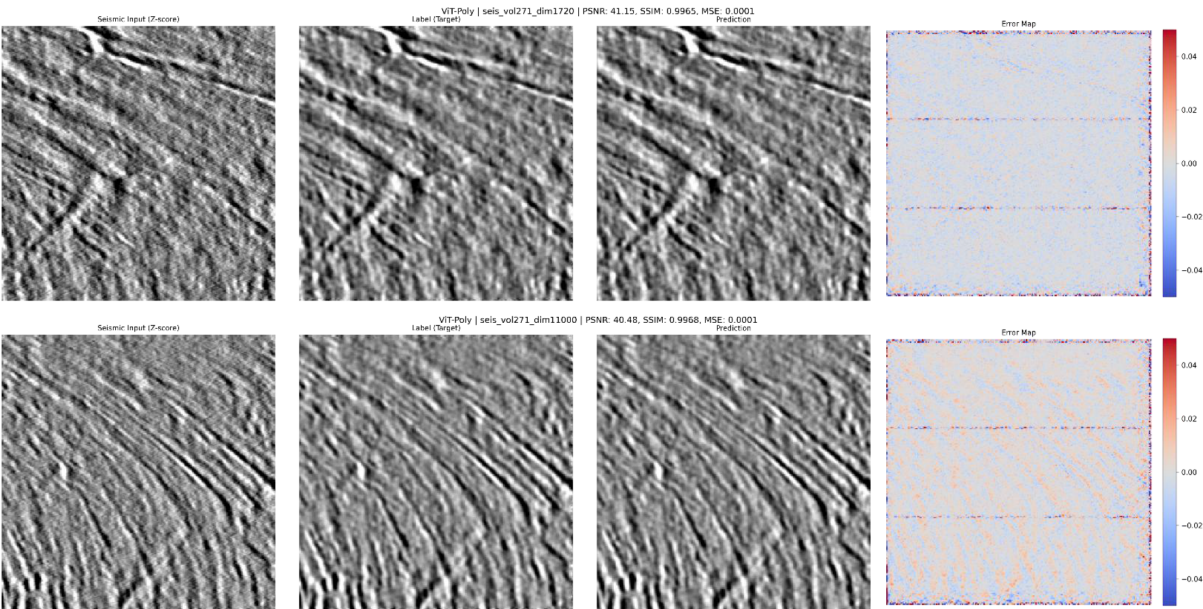


Figure 4—Visual comparison of Polynomial smoothing. Each row shows a validation patch with (left to right): raw input, Polynomial smoothed label, ViT-Poly output, and residual (ViT-Poly - Polynomial label). ViT predictions exhibit high-fidelity emulation, preserving geologic curvature and reflector continuity with minimal error.

Visually, the ViT-Poly model achieves near-identical reconstructions of the filtered targets. Reflector continuity, amplitude gradients, and thin-bed layering are precisely retained, even in areas with curvature or fault proximity. Compared to earlier tasks, the residual maps are the least textured, with errors largely constrained within ± 0.01 —indicating near-perfect structural and amplitude matching.

Quantitative results in Table 4 confirm this. ViT-Poly achieves a peak signal-to-noise ratio (PSNR) of 36.56 dB, structural similarity index (SSIM) of 0.9955, and mean squared error (MSE) of 0.0003. These values represent a significant improvement over the raw-to-filtered baseline (22.41 dB PSNR, 0.8458 SSIM, 0.0065 MSE), reflecting over 90% reduction in error.

Table 4—Quantitative performance of ViT-Poly against industry-standard filters EPS on validation data.

Method	PSNR (dB)	SSIM	MSE
Polynomial filter	22.41	0.8458	0.0065
Vit-Poly	36.56	0.9955	0.0003

Training Convergence

Fig. 5 illustrates the training loss curves for all three ViT variants. ViT-Random shows a steep initial drop from ~ 0.6 to 0.2 within 20 epochs, followed by gradual stabilization. ViT-EPS converges faster, reaching

~0.1 by epoch 15, while ViT-Poly achieves the lowest final loss, plateauing below 0.09 within the first 10 epochs. All models exhibit smooth, monotonic behavior with slight fluctuations due to cosine learning rate decay. This demonstrates the efficiency of fine-tuning on top of a pretrained SFM encoder using the hybrid MSE and MS-SSIM loss defined in Section 3.3.

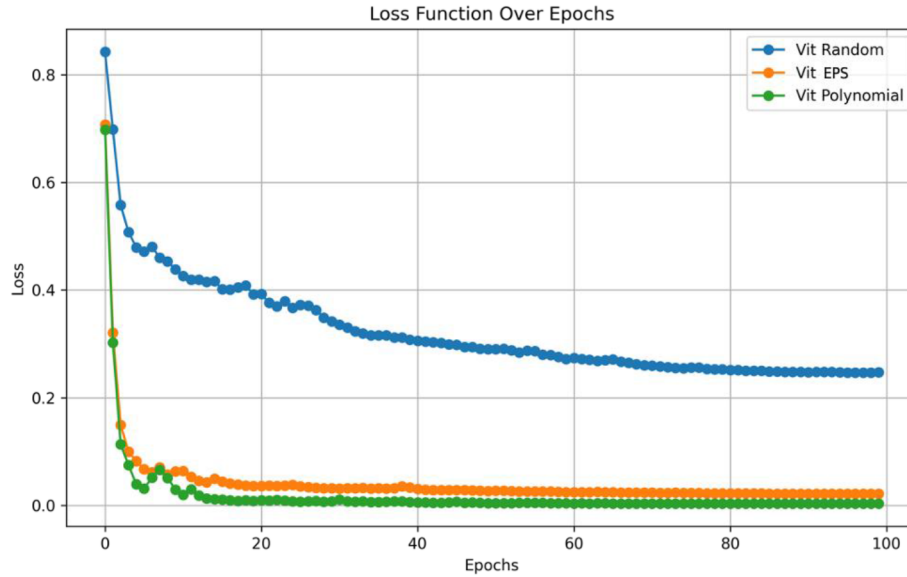


Figure 5—Fine-tuning loss curves for ViT-Random (blue), ViT-EPS (orange), and ViT-Poly (green). ViT-Poly converges fastest; ViT-Random exhibits a more gradual descent. All models stabilize within 30 epochs.

Evaluation Metrics

Model performance is assessed using three complementary metrics that quantify denoising fidelity at both pixel and structural levels:

- Mean Squared Error (MSE): Measures the average squared difference between the predicted and target pixel values. A lower MSE indicates higher pixel-wise accuracy.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- Peak Signal-to-Noise Ratio (PSNR): Expresses the reconstruction quality as a logarithmic ratio of the maximum signal power to the MSE. Higher values indicate better fidelity.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{R^2}{\text{MSE}} \right) \quad (\text{dB})$$

where R is the dynamic range of the input (typically 1.0 for normalized data).

- Structural Similarity Index Measure (SSIM): Proposed by Wang et al. (2004), it quantifies perceptual similarity by comparing luminance, contrast, and structure. A score of 1.0 indicates perfect structural alignment.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Where μ , σ^2 , σ_{xy} denote the mean, variance, and covariance of the predicted and reference images, and C_1 , C_2 are constants to stabilize the division.

These metrics are reported across all validation experiments and for inference on the unseen Norway F3 dataset.

Discussion

This study demonstrates the effectiveness of leveraging a pretrained Seismic Foundation Model (SFM) for seismic denoising through minimal supervised fine-tuning. The results affirm three key insights: the value of foundation model pretraining, the structural fidelity achieved by ViT-based outputs, and the generalization capability enabled by fine-tuning on synthetic random noise.

First, the pretrained SFM backbone enables efficient adaptation to diverse denoising tasks—random noise suppression, edge-preserving smoothing emulation, and polynomial smoothing emulation—using only a modest number of labeled seismic patches. The encoder, pretrained on over 2 million seismic patches from a broad range of geological settings, generalizes effectively even without data augmentation. The resulting training curves (Fig. 5) are smooth and monotonic across all tasks, reflecting stable optimization. ViT-Poly converges the fastest, likely due to the smoother statistical structure of its target outputs.

Second, ViT-based models demonstrate superior structural preservation compared to conventional filters. In both validation and generalization experiments, ViT-Random consistently recovers fault edges and stratigraphic continuity more effectively than median filtering. Likewise, ViT-EPS and ViT-Poly emulate their corresponding smoothing algorithms while better retaining fine-scale features—such as thin beds, channel boundaries, and minor reflectors—that are frequently lost in traditional filtering pipelines. These differences are clearly observed in the residual maps and visual comparisons (Figs. 1–4), which highlight reduced over-smoothing and improved geologic texture retention in ViT outputs.

Third, the generalization experiment on the unseen Norway F3 cube underscores the robustness of ViT-Random relative to EPS and polynomial smoothing. Although its quantitative metrics (e.g., SSIM and PSNR) are modestly lower than those of classical filters, the ViT-Random output preserves stratigraphic clarity and subtle discontinuities. Notably, this performance is achieved without any re-training or hyperparameter tuning. ViT-Random was fine-tuned solely on synthetic random noise, yet successfully adapted to real field data—suggesting that carefully designed noise injection during supervision can yield models that are inherently transferable to real-world seismic scenarios.

Despite these strengths, several limitations remain. The ViT models operate on 2D slices, potentially limiting their capacity to capture long-range 3D geologic context. Furthermore, while inference is efficient, interpretability and broader field validation remain important directions for future deployment. This study also focuses on random noise; further research is needed to extend robustness to other domain-specific artifacts, including coherent noise, aliasing, and acquisition-related distortions.

Overall, this work makes a compelling case for fine-tuning pretrained foundation models in seismic interpretation. ViT-MAE architectures deliver accurate, structure-preserving denoising and enable flexible deployment across tasks with minimal supervision—bridging the gap between data-driven learning and geophysical domain expertise.

Conclusion

This study introduced a practical and adaptable framework for seismic denoising by fine-tuning a pretrained Seismic Foundation Model (SFM) on task-specific objectives with minimal labeled data. Three Vision Transformer-based models—ViT-Random, ViT-EPS, and ViT-Poly—were developed to suppress random noise and emulate classical filters, including edge-preserving and polynomial smoothing techniques.

Quantitative and qualitative results demonstrate that these fine-tuned models outperform or closely match traditional methods while offering superior preservation of critical structural features such as faults, thin beds, and stratigraphic discontinuities. In particular, the ViT-Random model, trained solely on synthetic

perturbations, generalized effectively to unseen real-world data (e.g., the Norway F3 volume) without retraining—highlighting the robustness of foundation models for geophysical applications.

These findings underscore the potential of vision transformers not only as powerful denoisers but also as flexible emulators of domain-standard filtering methods. The proposed approach bridges the gap between classical geophysical practices and modern deep learning, offering a scalable and structure-aware solution for seismic interpretation workflows.

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